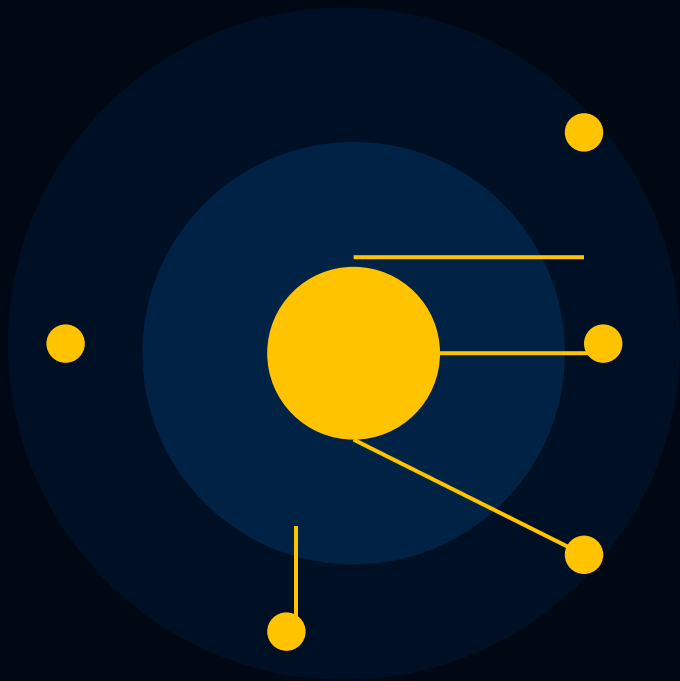


SmolVLM: Redefining Small and Efficient Multimodal Models

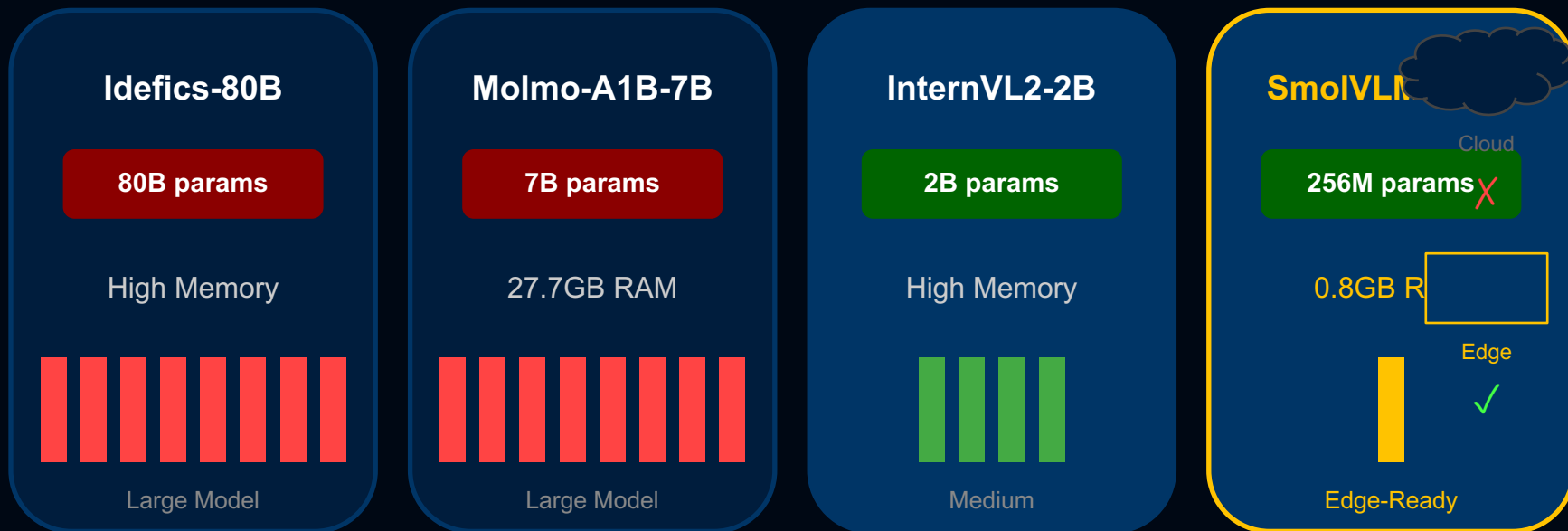
Hugging Face & Stanford University

[arXiv:2504.05299](https://arxiv.org/abs/2504.05299)



Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art VLMs require massive compute and memory, limiting mobile/edge deployment



300× smaller than Idefics-80B

Key Insight: Model size reduction enables edge deployment without significant accuracy loss

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone

1

Image Splitting

Images split into sub-images

2

Vision Encoder

SigLIP encoder processes sub-images

3

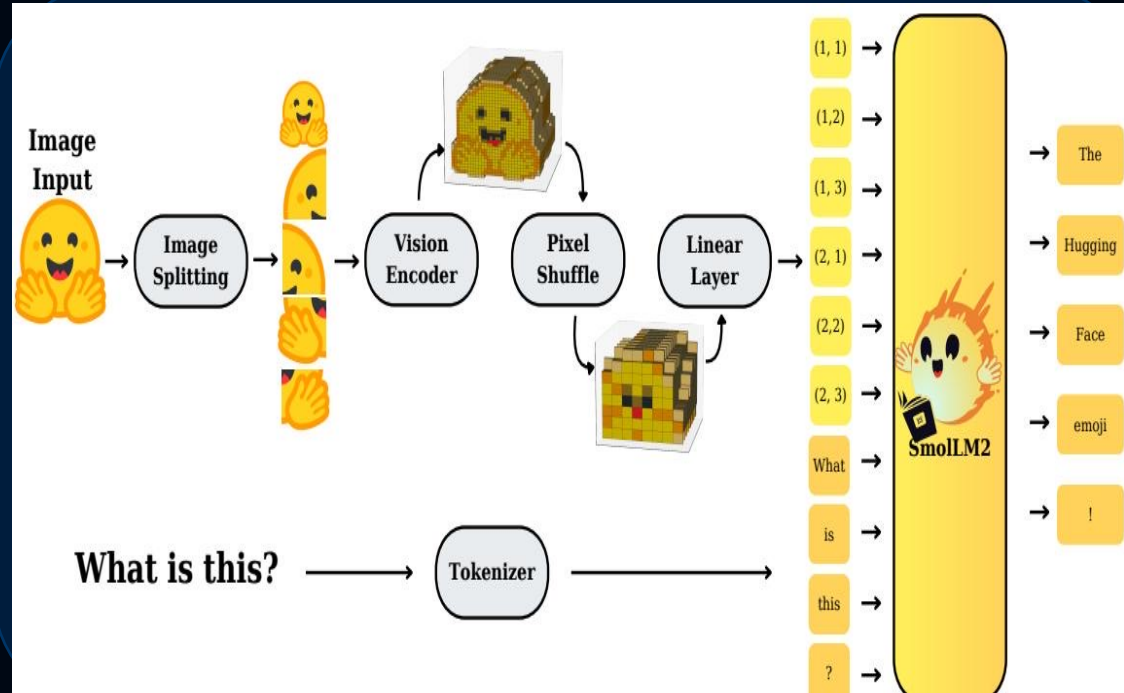
Pixel Shuffle

Space-to-depth compression

4

Linear Projection

Project into language model input space

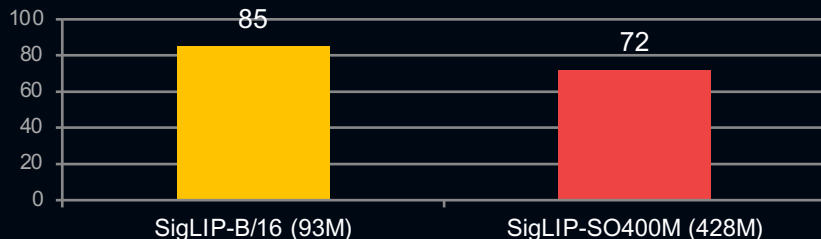
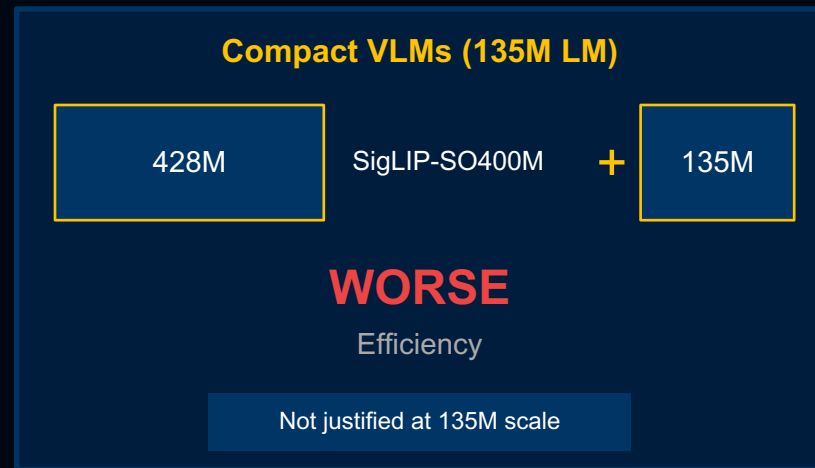
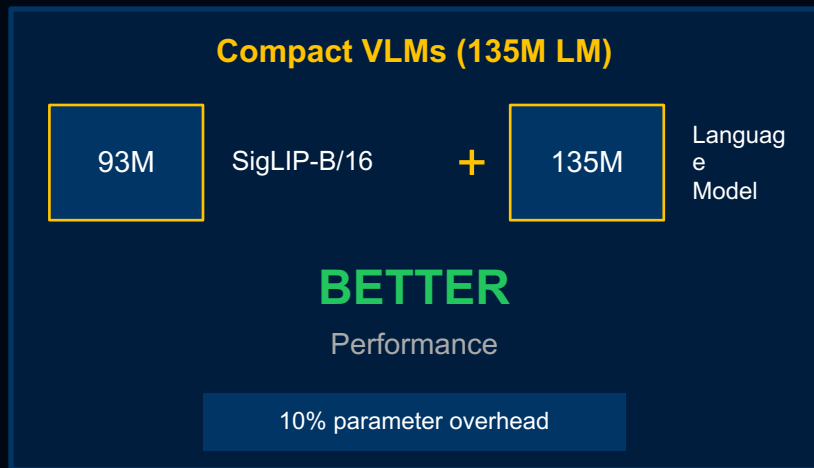


SmolVLM Processing Pipeline

Key: SigLIP encoder + Pixel Shuffle + SmolLM2 = Efficient Edge VLM

SmolVLM: Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1: Compute Allocation



KEY INSIGHT

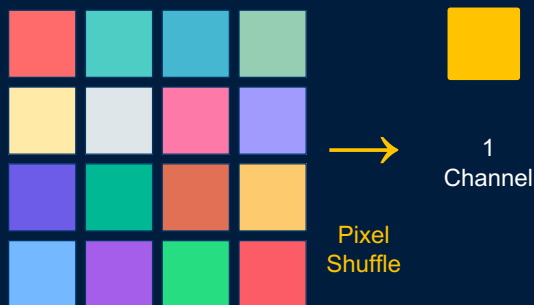
Balanced encoder-LM parameter allocation matters more than raw encoder size.

Only at 1.7B LM scale does the larger encoder (428M) justify its 10% parameter overhead.

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Finding 2: Pixel Shuffle Mechanism

Pixel Shuffle: Spatial → Channel Rearrangement



Tokens Reduced = r^2 ($r=4 \rightarrow 16\times$ reduction)

KEY BENEFIT FOR SMALL VLMs

Eases attention overhead by cutting tokens 16×, enabling efficient processing without sacrificing quality.

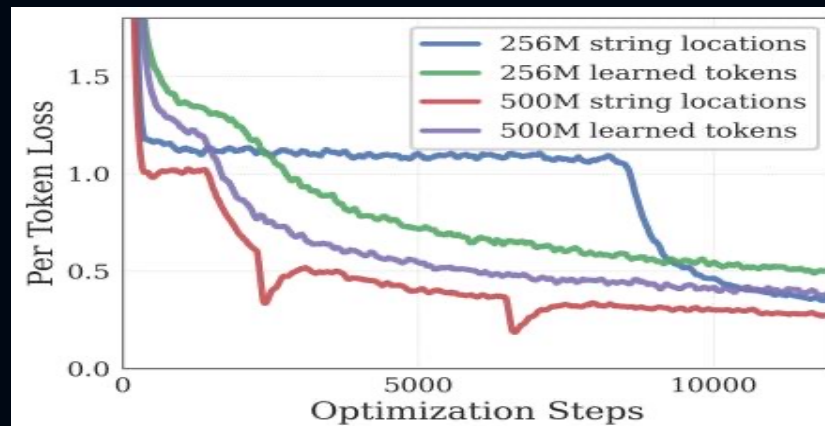


Figure 6: Pixel shuffle visualization

$r = 2$

4× reduction

Large models

$r = 4$

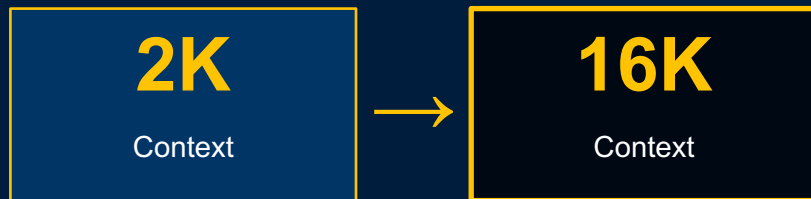
16× reduction

Small VLMs

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2: Context Length Scaling

Context Length Extension



RoPE Base Adjustment

10K → 273K

Fine-tuning on mixed long/short-context data

Performance improves significantly with extended context

Model Context Variants

2.2B Model

16K Context

Full capability for high-res image processing

256M / 500M Models

8K Context

Optimized for efficiency with smaller scale

KEY BENEFIT

Unlocks Higher Image Resolution

Longer context enables processing of larger images with more visual details

SmoIVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Benchmark Performance Comparison

Model	Avg Score	RAM Usage
SmoIVLM 256M	44.0%	0.8GB
SmoIVLM 500M	51.0%	1.2GB
SmoIVLM 2.2B	59.8%	4.9GB
Molmo-A1B-7B	(baseline)	27.7GB

Key Highlights

- SmoIVLM-256M surpasses 300× larger Llafix-80B
- RAM usage: 0.8GB to 4.9GB vs 27.7GB
- Up to 34× less RAM than Molmo-A1B-7B

Benchmark Categories

Single-Image

OCRBench, AI2D, ChartQA, TextVQA, DocVQA, ScienceQA

Multi-task

MMMU, MathVista, MMStar

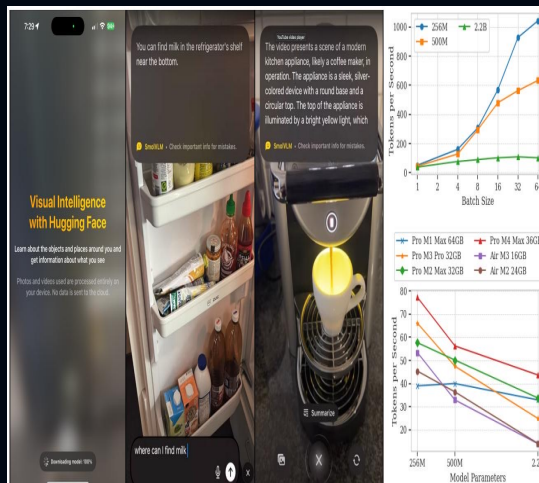
Video

Video-MME, MLVU

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon

On-Device Deployment

- HuggingSnap iOS app processes prompts entirely locally (no cloud)
- Throughput scales linearly with batch size for the smallest model



HuggingSnap iOS App

Throughput Benchmarks Across Apple Silicon Devices

~80

tokens/sec

SmolVLM-256M on Pro
M1 Max 64GB

~55

tokens/sec

MacBook Air M3 16GB

~1000

tokens/sec

Batch size 64 on high-end
devices

Throughput scales linearly with batch size for smallest model

Key Takeaways: Design Principles for Efficient Multimodal Models

- ✓ Balance encoder-LM compute: smaller encoders outperform larger ones for compact VLMs
- ✓ Extend context length: 16K tokens unlocks higher image resolution
- ✓ Use aggressive token compression: $r=4$ (16× reduction) for small models
- ✓ Sub-1GB inference is achievable: SmolVLM-256M runs at 0.8GB
- ✓ Fully open-source: All models, data, code, and mobile app released

SmolVLM redefines what's possible for on-device multimodal AI